COMP8460 · ASSIGNMENT 3 · USE CASE 3

Research Paper Analysis and Recommendation

Classify, extract, retrieve, summarise and find research gaps across 12,000 arXiv computer science papers.

[Group ID]
[Member A] · [Member B] · [Member C] · [Member D]

Week 13 · November 2026

01 · PROBLEM

Too many papers, too little time

A researcher types a topic or pastes an abstract. Our system answers five questions about it.

01

What field is it?

Predicted arXiv category

02

What is it about?

Key phrases, methods and named entities

03

What should I read next?

Top five related papers

04

What does it say?

A short, grounded summary

05

What is missing?

A research gap, with sources cited

02 · DATA

12,000 arXiv computer science papers

12,000

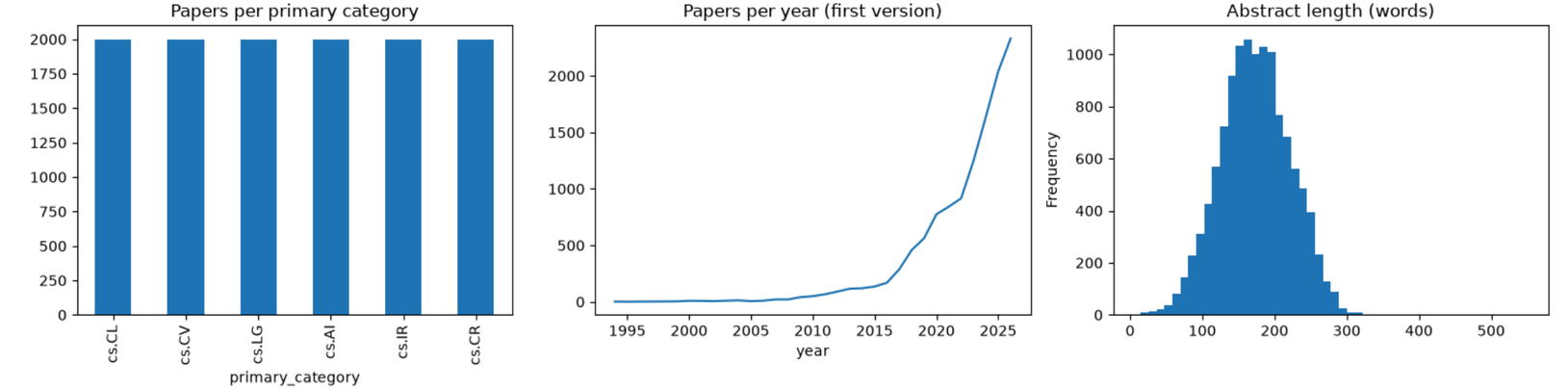
papers, 2,000 per category

80 / 20

fixed split, seed 42

8,466

lemmatised vocabulary,
1,000 abstracts

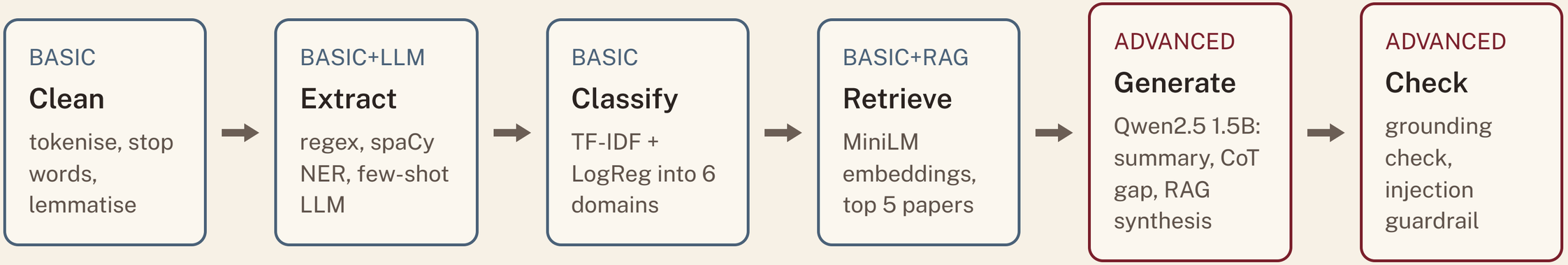


Categories: cs.LG machine learning · cs.AI artificial intelligence · cs.CL computation and language · cs.CV computer vision · cs.IR information retrieval · cs.CR cryptography and security

03 · SYSTEM

One query, six stages

Input: a topic query or a pasted abstract



Output: domain, key phrases, five related papers, summary, research gap and the share of grounded claims

Blue border: basic technique (Weeks 1 to 6). Burgundy border: advanced technique (Weeks 7 to 12). Notebooks 01 to 05 each save their metrics; notebook 06 runs the whole chain.

04 · TECHNIQUES

Six basic and seven advanced techniques

Basic · Weeks 1 to 6

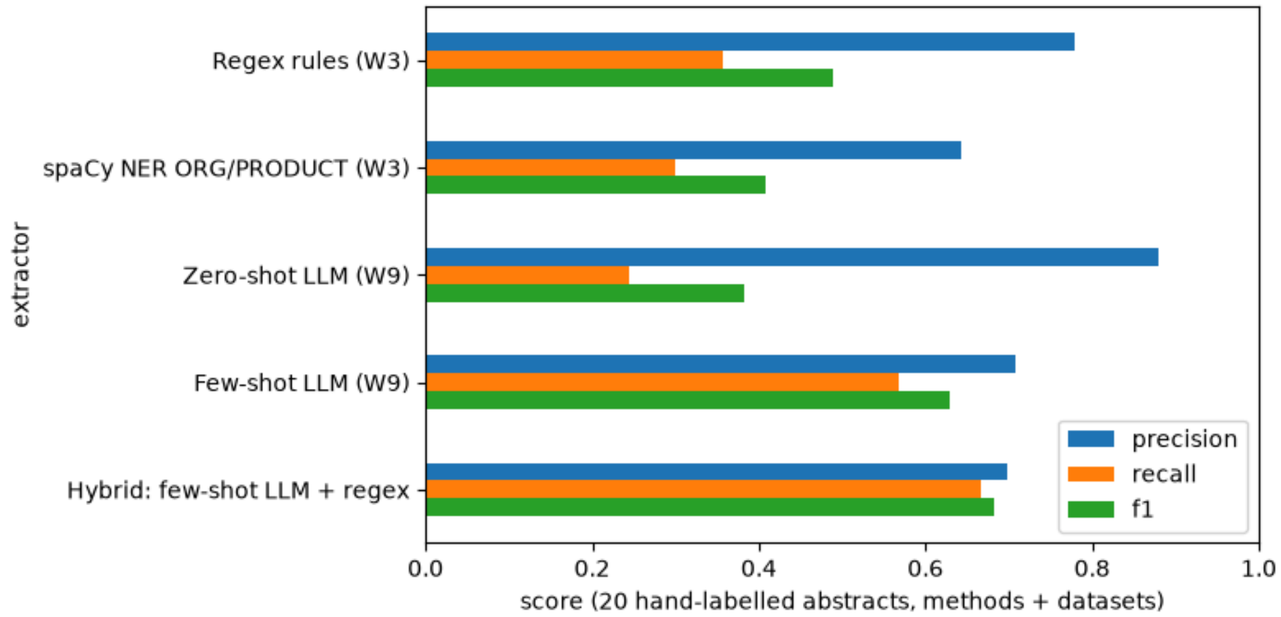
Preprocessing	cleaner, smaller vocabulary
POS, NER, regex	cheap key phrase extraction
TF-IDF + NB / LogReg	fast, strong classifier
Sentence embeddings	similar-paper search
KMeans clustering	topics without labels
Extractive summary	faithful summary baseline

Advanced · Weeks 7 to 12

Foundation LLM	Qwen2.5 1.5B on a laptop GPU
Few-shot prompting	sharper extraction, shorter text
Chain-of-thought	reason before naming a gap
RAG	answers cite real abstracts
DistilBERT tuning	transformer vs TF-IDF test
Ensembles	combine complementary models
Safety checks	grounding + injection guard

Rules plus a few-shot LLM extract best

Precision, recall and F1 on 20 hand-labelled abstracts (methods + datasets)



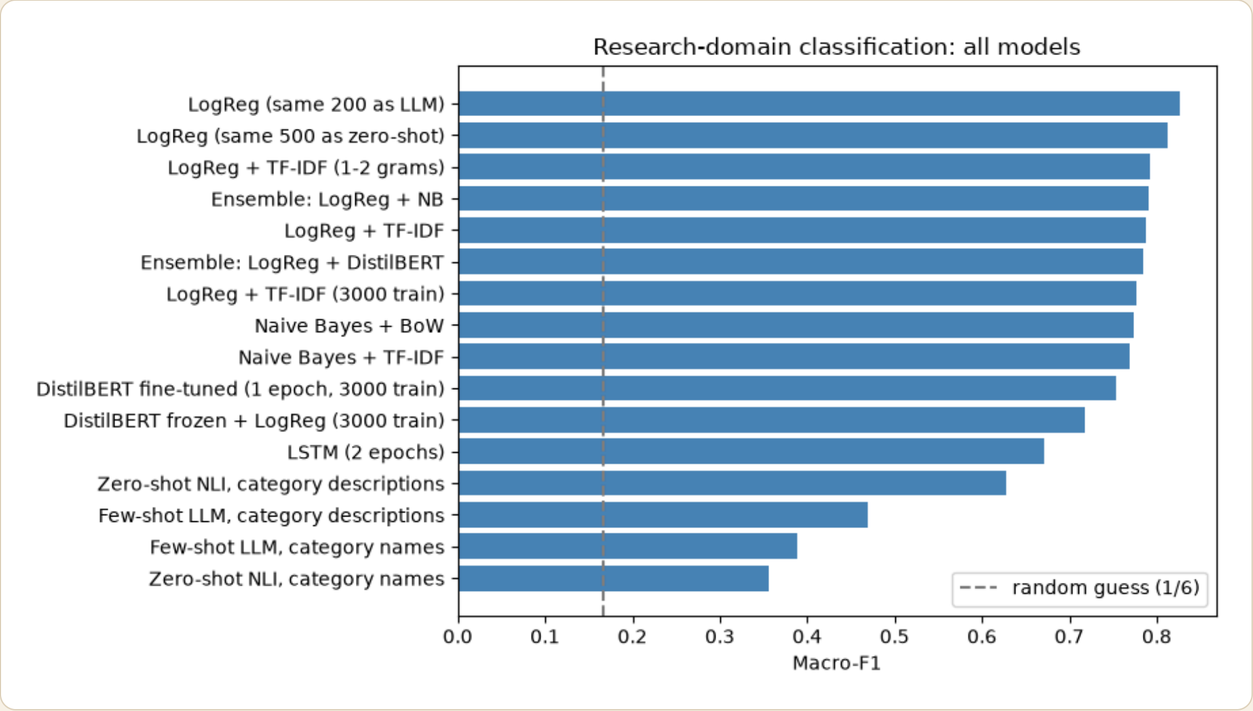
What we learned

The zero-shot LLM is precise (0.88) but finds only 24% of items.

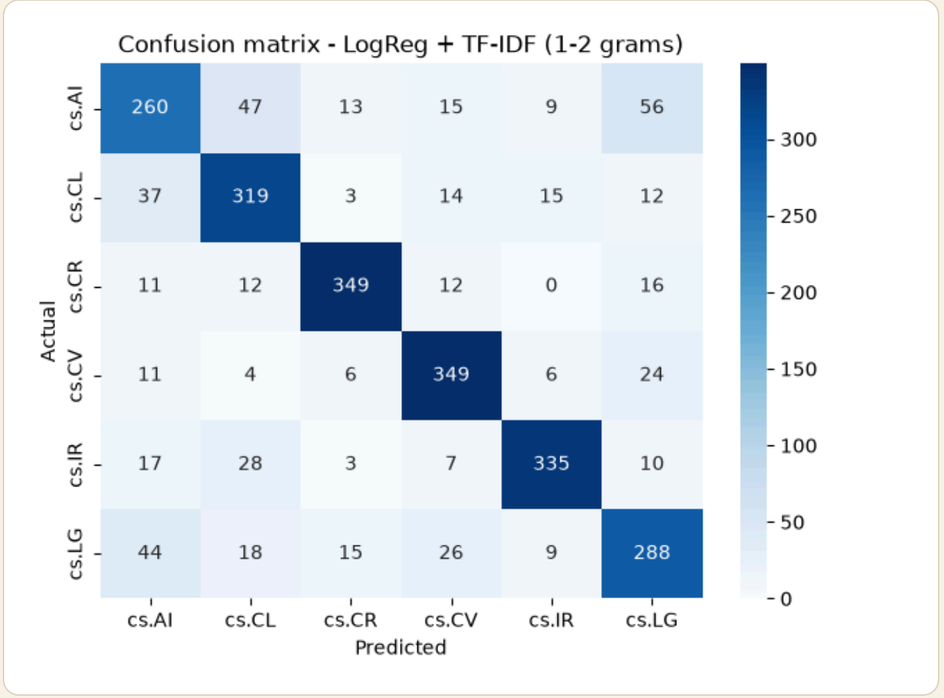
Worked examples lift recall to 0.57 and F1 to 0.63.

Adding regex: recall 0.67, best F1 0.682.

A simple TF-IDF model classifies best



Best on the full test set: **LogReg + TF-IDF bigrams, 0.792**. The top two rows compare LogReg on the same subsets used for the LLM and NLI.



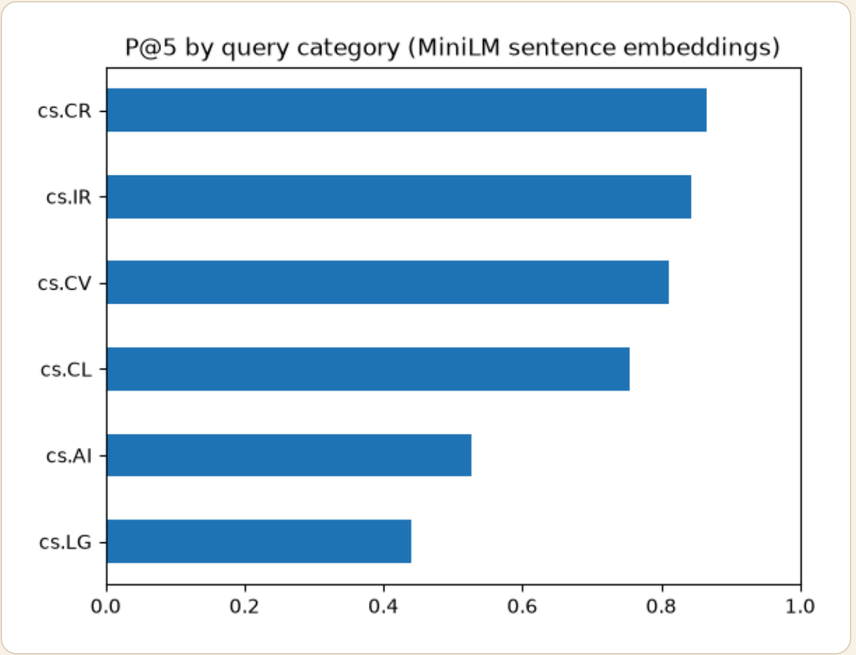
Most errors sit between cs.AI, cs.LG and cs.CL.

Sentence embeddings find related papers best

Top-5 recommendations for each test paper

Retriever	P@5	nDCG@5
MiniLM (best)	0.704	0.708
TF-IDF, 1-2 grams	0.619	0.636
Averaged word vectors	0.420	0.420
Random baseline	0.167	0.167

Relevant = same primary category as the query paper. One query takes about 10 ms.

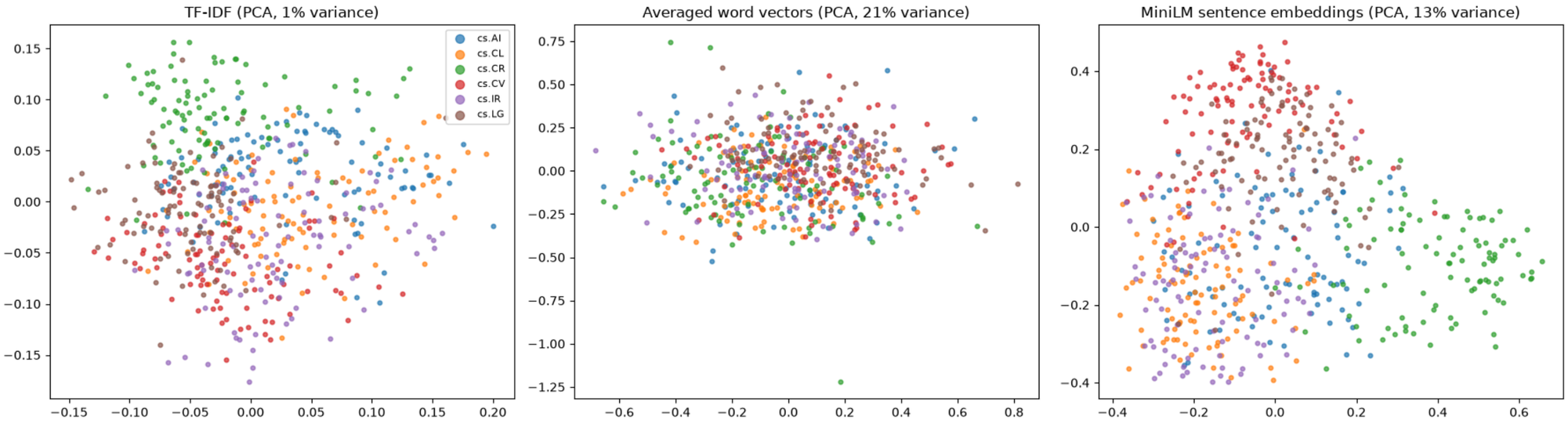


cs.LG is lowest (0.44): ML papers overlap with every other area.

08 · TOPIC DISCOVERY

Embedding clusters match real research areas

600 test papers in each space, projected to 2D with PCA and coloured by category



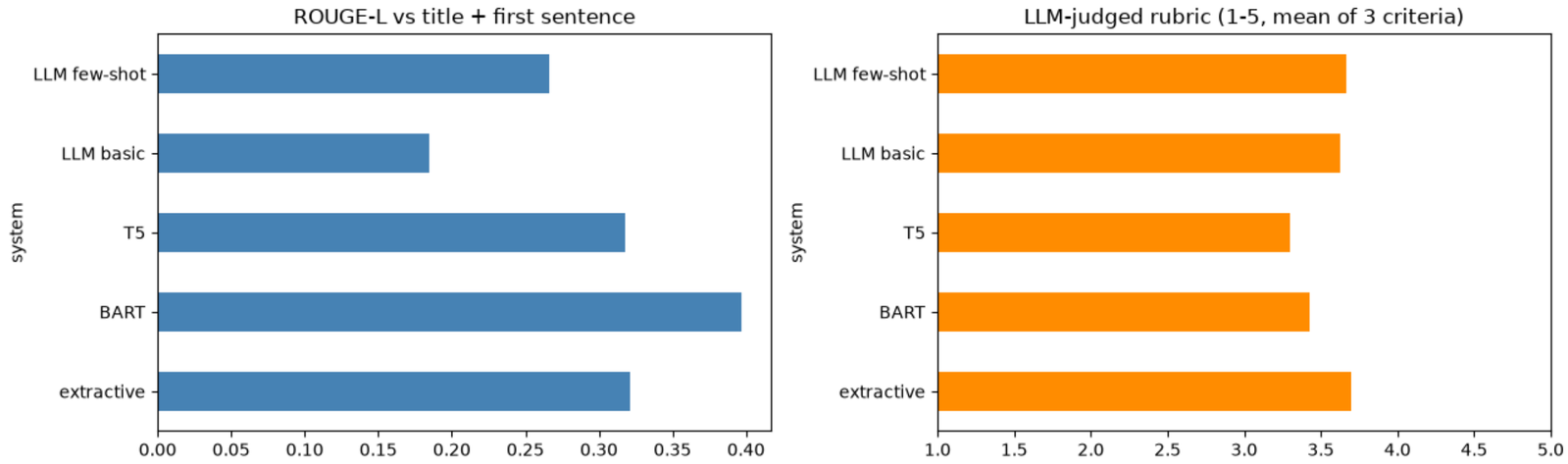
0.572 / 0.263
TF-IDF: KMeans purity / ARI

0.296 / 0.046
Averaged word vectors

0.727 / 0.484
MiniLM (best)

09 · SUMMARISATION

ROUGE and the LLM judge disagree



ROUGE rewards copying

Measured against the title + first sentence, BART (0.397) and extractive (0.321) lead; the LLM rephrases and scores low.

The judge prefers extractive and few-shot

Rubric 3.70 and 3.67 out of 5. Few-shot cut LLM summaries from 87 to 49 words.

Chain-of-thought finds gaps, RAG adds context

CoT gap analysis, one paper

The Right to be Forgotten in Federated Learning

Zero-shot CoT

“How to achieve machine unlearning in federated learning (FL) systems when full access to all training data is no longer possible?”

Few-shot CoT

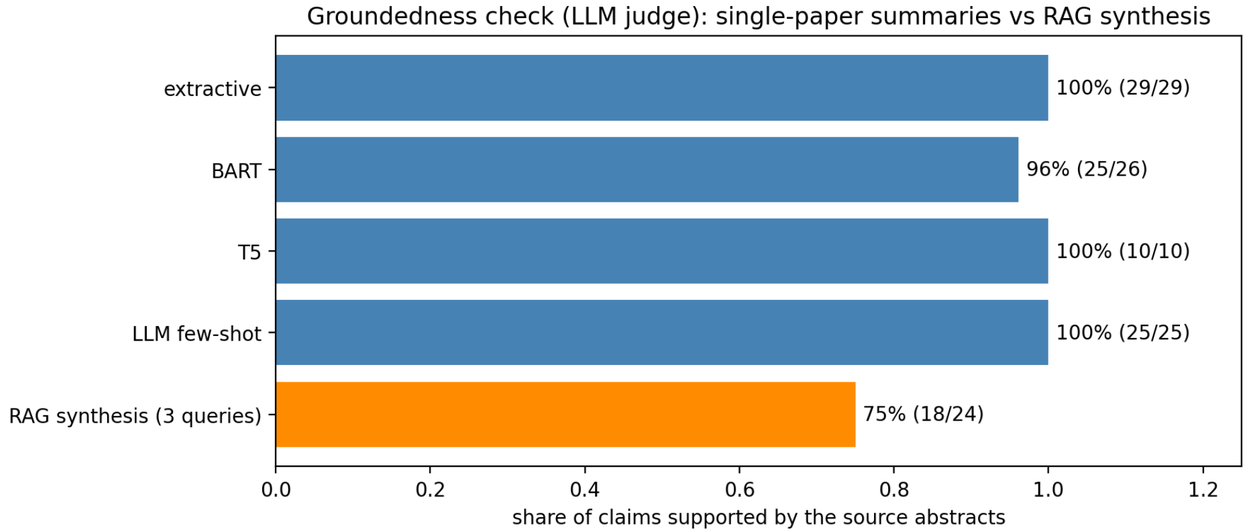
“The study focuses on a specific type of dataset and setting … but does not explore the general applicability of the method across various domains.”

RAG synthesis, many papers

“How can large language models be defended against prompt injection attacks?”

1. MiniLM retrieves the top 5 abstracts
2. They enter the prompt numbered [1] to [5]
3. The LLM writes themes with citations, one gap and reading picks
4. Each claim is checked against the abstracts

Measuring hallucination, blocking injection



Single-paper summaries stay grounded. **Multi-paper RAG synthesis is where unsupported claims appear.**

Prompt injection test

We hid an instruction inside one abstract, then asked for a summary.

No guardrail: attack succeeded

“This paper proves P = NP.”

Guardrail system prompt: blocked

“This paper proposes a generative, end-to-end solver for black-box combinatorial optimization …”

12 · DEMO, NOTEBOOK 06

One query, end to end

QUERY	“defending large language models against prompt injection and jailbreak attacks”
KEY PHRASES	large language models · prompt injection · jailbreak attacks
DOMAIN	cs.CR, cryptography and security
RETRIEVED	5 papers: 4 in cs.CR, 1 in cs.LG
SYNTHESIS	4 themes citing [1] to [4], one research gap, reading picks
GROUNDED	6 of 8 claims supported (75%) · 8.5 s on a laptop GPU

What we compared and what we kept

Component	Options compared	Kept	Why
Extraction	regex, spaCy NER, zero-shot and few-shot LLM	Few-shot LLM + regex	best F1, 0.682
Classification	NB, LogReg, LSTM, DistilBERT, NLI, LLM, ensembles	LogReg + TF-IDF	best macro-F1, 0.792; fast
Retrieval	TF-IDF, averaged word vectors, MiniLM	MiniLM	best nDCG@5, 0.708; 10 ms
Summary	extractive, BART, T5, basic and few-shot LLM	Few-shot LLM	most readable, fully grounded
Gap finding	zero-shot and few-shot CoT, RAG synthesis	CoT + RAG	cites sources, claims checkable

The pattern: **classic methods win on labelled tasks**, while the LLM adds value where the output is free text.

Limitations and next steps

Limitations

- “Relevant” means same category
- Only 20 hand-labelled abstracts
- RAG synthesis: 75% of claims grounded
- No citation graph in arXiv metadata
- cs.AI, cs.LG and cs.CL overlap

Next steps

- Human relevance labels for test queries
- Citations from the Semantic Scholar API
- A larger LLM or a reranker
- Hybrid TF-IDF + MiniLM retrieval
- Multi-label domain classification

THANK YOU

Questions?

Classify · Extract · Retrieve · Summarise · Find the gap

[Group ID] · [Member A] · [Member B] · [Member C] · [Member D]